

palimpsestr: User Manual

Complete Reference Guide v0.10.0

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1 Introduction

palimpsestr is an R package for the probabilistic decomposition of archaeological palimpsests using the Stratigraphic Entanglement Field (SEF) framework. It integrates spatial proximity, vertical distribution, chronological overlap, and cultural similarity to estimate latent depositional phases.

1.1 Installation

```
# From GitHub
devtools::install_github("enzococca/palimpsestr")

# From local source
install.packages("palimpsestr_0.10.0.tar.gz", repos = NULL, type = "source")
```

1.2 Required and optional packages

	Package	Status	Purpose
stats, utils, graphics, grDevices	Required		Core R
ggplot2	Suggested		Publication-quality plots (<code>gg_*</code> functions)
plotly	Suggested		Interactive visualization (<code>as_plotly()</code>)
sf	Suggested		GIS export (<code>as_sf_phase()</code> , <code>as_sf_links()</code>)
DBI, RSQLite, RPostgres	Suggested		Database import (<code>read_db()</code>)

2 Data Preparation

2.1 Required columns

Every dataset must contain:

Column	Type	Description
x	numeric	Easting coordinate (metres)
y	numeric	Northing coordinate (metres)
z	numeric	Depth or elevation (metres)
date_min	numeric	Start of chronological interval
date_max	numeric	End of chronological interval
class	character	Material class (ceramic, lithic, bone, metal, etc.)

2.2 Optional columns

	Column	Type	Description
<code>id</code>	any		Unique identifier for each find
<code>context</code>	character		Stratigraphic unit label (e.g., “SU_1”)
<code>taf_score</code>	numeric (0–1)		Taphonomic disturbance score
<code>true_phase</code>	integer		Known phase (for validation only)

2.3 Notes on Z coordinates

Each artifact needs its own z value. If you don’t have point-level depth measurements, you can assign the mean depth of the stratigraphic unit (US) to all finds within it:

```
# Example: assign US mean depth to finds
data$z <- ave(data$z_us, data$context, FUN = mean)
```

2.4 Simulating test data

```
# Well-separated phases (low mixing)
easy <- archaeo_sim(n = 150, k = 3, seed = 1, mixing = 0.05)

# Moderate palimpsest
moderate <- archaeo_sim(n = 200, k = 3, seed = 1, mixing = 0.30)

# Compressed palimpsest (hard to resolve)
compressed <- archaeo_sim(n = 250, k = 4, seed = 1, mixing = 0.50)
```

Parameters:

- `n`: number of artifacts
- `k`: number of phases
- `seed`: random seed for reproducibility
- `mixing`: proportion of inter-phase mixing (0 = none, 1 = complete)

2.5 Built-in datasets

The package includes four built-in datasets:

Dataset	Finds	Phases	Mixing	Description
<code>demo_easy</code>	150	3	5%	Well-separated phases
<code>demo_moderate</code>	200	3	30%	Moderate palimpsest
<code>demo_compressed</code>	250	4	50%	Heavily disturbed
<code>villa_romana</code>	615	–	Real data	Poggio Gramignano Roman villa (VRPG 3004)

```

data(villa_romana)
str(villa_romana)
#> 'data.frame': 615 obs. of 9 variables:
#> $ id : chr "VRPG_0001" "VRPG_0002" "VRPG_0003" "VRPG_0004" ...
#> $ x : num 2299041 2299041 2299041 2299041 2299041 ...
#> $ y : num 4714126 4714126 4714126 4714126 4714126 ...
#> $ z : num 284 284 284 284 284 ...
#> $ context : chr "US_102" "US_102" "US_102" "US_102" ...
#> $ date_min : num 425 425 425 425 425 425 425 425 425 ...
#> $ date_max : num 450 450 450 450 450 450 450 450 450 ...
#> $ class : chr "Reperto Ceramico" "Reperto Ceramico" "Reperto Ceramico" "Reperto Ceramico" ...
#> $ taf_score: num 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 ...
#> - attr(*, "source")= chr "Poggio Gramignano (VRPG 3004), pyArchInit database"
#> - attr(*, "taf_source")= chr "Compiled by Roberto Montagnetti (April 2026)"
table(villa_romana$class)
#>
#> Campionature Intonaco Dipinto Materiale Da Costruzione
#> 5 1 38
#> Materiale Lapideo Pietre Dure-Gemme Reperti Antropologici
#> 1 1 1
#> Reperti Archeozoologici Reperto Anforaceo Reperto Ceramico
#> 7 76 417
#> Reperto Faunistico Reperto Lapideo Reperto Metallico
#> 1 6 19
#> Reperto Numismatico Reperto Vitreo Resti Di Pavimentazioni
#> 10 25 1
#> Scarti Di Produzione Stucco
#> 5 1
# Core Analysis

## Fitting the SEF model: `fit_sef()`

```

```

fit <- fit_sef(
  data      = moderate,
  coords    = c("x", "y", "z"),      # coordinate columns
  chrono    = c("date_min", "date_max"), # chronological range columns
  class     = "class",                # material class column
  tafonomy  = "taf_score",            # optional: taphonomic score
  context   = "context",              # optional: stratigraphic unit
  harris    = NULL,                   # optional: Harris Matrix penalty
  k         = 3,                      # number of phases
  weights   = c(ws = 1, wz = 1, wt = 1, wc = 1), # domain weights
  seed      = 42,                     # random seed
  em_iter   = 100,                    # max EM iterations
  em_tol    = 1e-5,                   # convergence tolerance
  n_init    = 5                       # number of random initialisations
)
print(fit)
#> <sef_fit>
#> Observations: 200
#> Estimated phases: 3
#> Mean entropy: 0.000
#> Mean local SEI: 117.643

```

```
#> Mean energy: 17.550
#> LogLik: -1416.627
#> BIC: 3129.960
#> ICL: 3129.920
#> PDI: 1.000
#> Converged: yes
#> Initialisations: 5
```

2.5.1 Parameters explained

Parameter	Default	Description
<code>k</code>	3	Number of depositional phases to estimate
<code>weights</code>	<code>c(ws=1, wz=1, wt=1, wc=1)</code>	Relative weights for spatial (ws), vertical (wz), temporal (wt), and cultural (wc) evidence
<code>em_iter</code>	100	Maximum EM iterations
<code>em_tol</code>	1e-5	Convergence tolerance (log-likelihood change)
<code>n_init</code>	1	Number of random initialisations (higher = more robust, slower)
<code>tafonomy</code>	NULL	Column name for taphonomic disturbance (0–1)
<code>context</code>	NULL	Column name for stratigraphic unit labels
<code>harris</code>	NULL	$n \times n$ penalty matrix (from <code>harris_from_contexts()</code> or <code>read_harris()</code>)

2.5.2 Output: the `sef_fit` object

The returned object contains:

Field	Type	Description
<code>\$phase</code>	integer vector	Dominant phase assignment per find
<code>\$phase_prob</code>	matrix	$n \times k$ matrix of phase membership probabilities
<code>\$entropy</code>	numeric vector	Shannon entropy per find
<code>\$energy</code>	numeric vector	ESE per find
<code>\$local_sei</code>	numeric vector	Local SEI per find
<code>\$converged</code>	logical	Whether EM converged
<code>\$n_init</code>	integer	Number of initialisations used
<code>\$em_loglik</code>	numeric vector	Log-likelihood trace across iterations
<code>\$model_stats</code>	list	Summary statistics (PDI, BIC, loglik, etc.)

2.6 Phase count selection: `compare_k()`

When K is unknown, compare multiple values:

```
ck <- compare_k(moderate, k_values = 2:5, seed = 42,
                tafonomy = "taf_score", context = "context")
print(ck)
#>   k      pdi mean_entropy mean_local_sei mean_energy      loglik      bic
#> 1 2 0.9929850 4.862417e-03      117.6428      17.55018 -1661.83619 3519.7101
#> 2 3 0.9999074 1.017010e-04      117.6428      17.55018 -1416.62731 3129.9604
#> 3 4 0.9999932 9.441871e-06      117.6428      17.55018  -588.74176 1574.8573
#> 4 5 0.9989364 1.711724e-03      117.6428      17.55018  -28.34952  554.7409
#>      icl tot_withinss pseudo_bic
#> 1 3517.7652      749.9725    359.7135
#> 2 3129.9197      545.5200    343.7390
#> 3 1574.8536      493.7309    371.4741
#> 4  554.0562      447.6697    399.5720
```

The optimal K minimises BIC while maximising PDI.

2.7 Key diagnostics

```
# Palimpsest Dissolution Index (global separability, 0--1)
pdi(fit)
#> [1] 0.9999074

# Compact summary
sef_summary(fit)
#> $n
#> [1] 200
#>
#> $k
#> [1] 3
#>
#> $pdi
#> [1] 0.9999074
#>
#> $mean_entropy
#> [1] 0.000101701
#>
#> $mean_energy
#> [1] 17.55018
#>
#> $loglik
#> [1] -1416.627
#>
#> $bic
#> [1] 3129.96
#>
#> $phase_count
#>
#>   1  2  3
#> 66 68 66

# Phase probability table
```

```

head(as_phase_table(fit))
#>      id dominant_phase phase1      phase2      phase3      entropy local_sei
#> 1 F001              1      1 3.464056e-17 9.447792e-41 1.312929e-15 126.0263
#> 2 F002              1      1 6.500699e-17 4.337547e-41 2.422943e-15 118.9763
#> 3 F003              1      1 2.885892e-19 1.470422e-45 1.231967e-17 103.0079
#> 4 F004              1      1 2.313963e-21 1.196827e-47 1.099487e-19 125.0059
#> 5 F005              1      1 4.942410e-14 1.953203e-35 1.563788e-12 130.4531
#> 6 F006              1      1 2.007763e-16 1.345985e-42 7.478970e-15 134.3008
#>      energy
#> 1 23.49767
#> 2 14.49467
#> 3 15.29267
#> 4 17.01351
#> 5 19.62919
#> 6 16.21251

# Full diagnostic table (input + diagnostics)
head(phase_diagnostic_table(fit))
#>      id      x      y      z context date_min date_max      class taf_score
#> 1 F001 12.36888 104.53174 11.79768 SU_4 970.6746 1011.055 lithic 0.4687699
#> 2 F002 19.12233 89.86743 19.78304 SU_3 960.5955 1080.232 lithic 0.3796706
#> 3 F003 24.19310 96.13386 18.97572 SU_7 966.4167 1030.552 bone 0.2288079
#> 4 F004 26.50473 99.62853 18.50500 SU_2 947.1761 1027.755 lithic 0.2246840
#> 5 F005 45.78809 91.97095 17.58970 SU_1 975.1793 1071.124 ceramic 0.1925518
#> 6 F006 32.65961 89.87875 17.67439 SU_4 946.7372 1055.110 ceramic 0.5558153
#>      true_phase dominant_phase phase1      phase2      phase3      entropy
#> 1              1              1      1 3.464056e-17 9.447792e-41 1.312929e-15
#> 2              1              1      1 6.500699e-17 4.337547e-41 2.422943e-15
#> 3              1              1      1 2.885892e-19 1.470422e-45 1.231967e-17
#> 4              1              1      1 2.313963e-21 1.196827e-47 1.099487e-19
#> 5              1              1      1 4.942410e-14 1.953203e-35 1.563788e-12
#> 6              1              1      1 2.007763e-16 1.345985e-42 7.478970e-15
#>      local_sei      energy
#> 1 126.0263 23.49767
#> 2 118.9763 14.49467
#> 3 103.0079 15.29267
#> 4 125.0059 17.01351
#> 5 130.4531 19.62919
#> 6 134.3008 16.21251

```

3 Intrusion Detection

```

di <- detect_intrusions(fit)
suspects <- di[di$intrusion_prob > 0.5, ]
cat("Suspected intrusions:", nrow(suspects), "\n")
#> Suspected intrusions: 19
head(suspects[order(suspects$intrusion_prob, decreasing = TRUE), ])
#>      id intrusion_prob
#> 24 F024      0.5895724
#> 112 F112      0.5652644
#> 31 F031      0.5626667

```



```
#> 48 F048      0.5579712
#> 77 F077      0.5547944
#> 145 F145     0.5484063
```

4 Model Validation

When true phase labels are known:

```
# Adjusted Rand Index (-1 to 1, 1 = perfect)
adjusted_rand_index(fit, moderate$true_phase)
#> [1] 0.9848866

# Confusion matrix (auto-reordered for best diagonal match)
confusion_matrix(fit, moderate$true_phase)
#>           true
#> estimated  1  2  3
#>           1 66  0  0
#>           2  1 67  0
#>           3  0  0 66
```

5 Structured Exports

5.1 Summary per stratigraphic unit

```
us_summary_table(fit)
#>   context n_finds dominant_phase   purity mean_entropy mean_energy
#> 1  SU_1      18           1 0.3888889 1.128512e-03 17.13106
#> 2  SU_2      29           3 0.3793103 2.769511e-13 17.13525
#> 3  SU_3      20           1 0.3500000 5.692683e-11 18.87997
#> 4  SU_4      20           2 0.4500000 5.459267e-09 17.58672
#> 5  SU_5      16           2 0.3750000 6.878401e-13 17.74102
#> 6  SU_6      29           2 0.3793103 1.140044e-10 17.04508
#> 7  SU_7      23           3 0.4347826 6.975216e-10 18.04596
#> 8  SU_8      16           2 0.5625000 5.610695e-07 17.20768
#> 9  SU_9      29           1 0.4137931 6.163456e-07 17.47853
#>   mean_local_sei n_intrusions
#> 1      120.8741           3
#> 2      114.3377           2
#> 3      113.3714           5
#> 4      114.9627           1
#> 5      117.8229           1
#> 6      118.1819           2
#> 7      121.6252           3
#> 8      125.1586           0
#> 9      115.7931           2
```

Columns: context, n_finds, dominant_phase, purity, mean_entropy, mean_energy, mean_local_sei, n_intrusions.

5.2 Phase transition matrix

Shows how often finds from phase i sit directly above finds from phase j :

```
phase_transition_matrix(fit)
#>      phase1 phase2 phase3
#> phase1     59      7      0
#> phase2      4     28     35
#> phase3      2     33     31
```

5.3 Export all results to CSV

```
export_results(fit, dir = "results/", format = "csv", prefix = "site_X")
```

Creates 4 files:

- `site_X_phases.csv` — phase assignments and diagnostics per find
- `site_X_intrusions.csv` — intrusion probabilities
- `site_X_us_summary.csv` — per-US aggregated statistics
- `site_X_model_summary.csv` — model-level metrics (n, k, PDI, BIC, etc.)

6 Harris Matrix Integration

6.1 Auto-generate from depth ordering

```
H <- harris_from_contexts(moderate, z_col = "z", context_col = "context",
                          penalty_scale = 0.5)
dim(H)
#> [1] 200 200
```

6.2 Import from CSV

The CSV must have columns `from`, `to`, and optionally `weight`:

```
from,to,weight
SU_1,SU_2,1
SU_2,SU_3,1
SU_1,SU_3,0.5
```

```
H <- read_harris("harris_matrix.csv", contexts = data$context)
fit <- fit_sef(data, k = 3, harris = H, context = "context")
```

6.3 Validate phases against stratigraphy

```

validate_phases_harris(fit)
#>   upper lower upper_phase lower_phase consistent
#> 1  SU_1  SU_2           1           3      TRUE
#> 2  SU_2  SU_5           3           2     FALSE
#> 3  SU_5  SU_7           2           3      TRUE
#> 4  SU_7  SU_6           3           2     FALSE
#> 5  SU_6  SU_4           2           2      TRUE
#> 6  SU_4  SU_9           2           1     FALSE
#> 7  SU_9  SU_3           1           1      TRUE
#> 8  SU_3  SU_8           1           2      TRUE

```

The `consistent` column flags whether the dominant phase ordering respects stratigraphic depth. `FALSE` indicates an inversion that may warrant investigation.

7 Visualization

7.1 ggplot2 plots (static)

All `gg_*` functions require **ggplot2** and return ggplot objects.

```

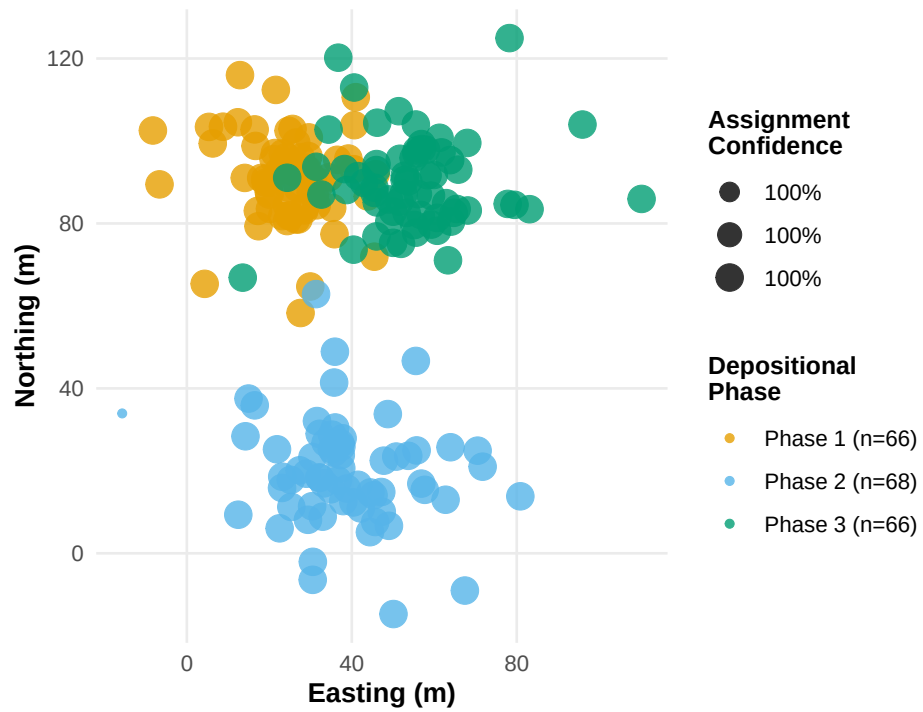
library(ggplot2)

# Phase assignment map
gg_phasefield(fit)

```

Dominant Phase Assignment

200 finds assigned to 3 depositional phases | PDI = 1.000



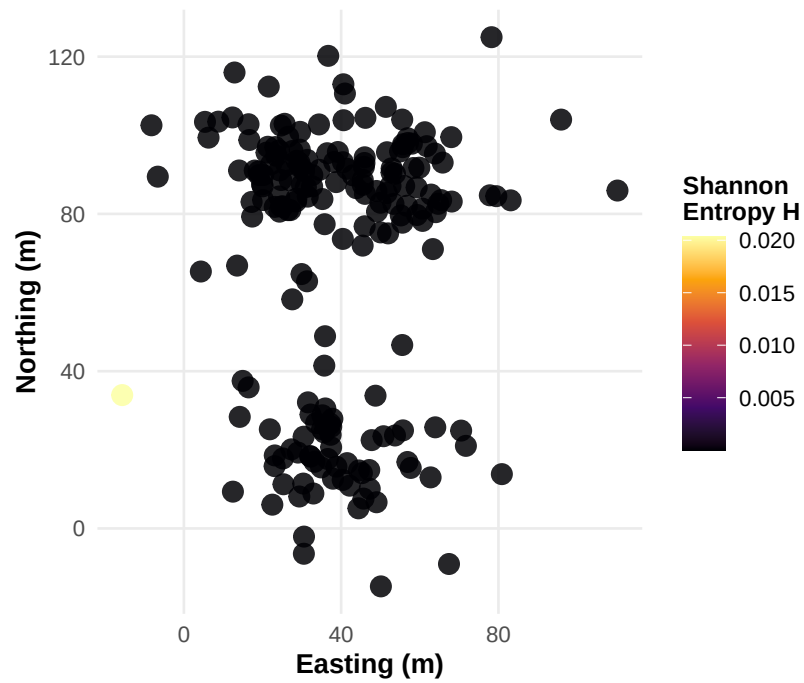
Generated by palimpsestr 0.12.0

```
# Entropy map (uncertainty)  
gg_entropy(fit)
```

Spatial Entropy Map

Entropy measures uncertainty in phase assignment

Dark = confident assignment | Bright = ambiguous (possible mixing zone)

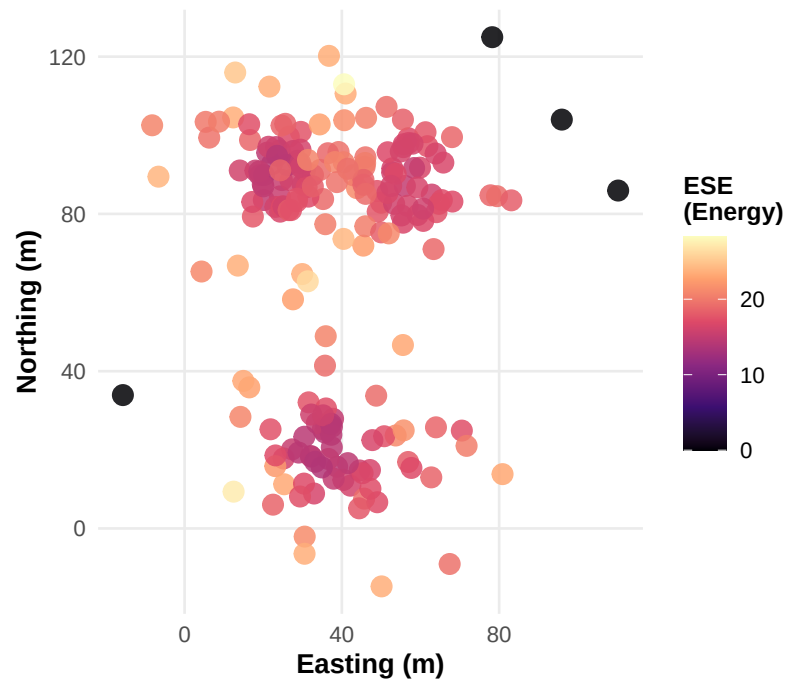


```
# Energy map (local disruption)
gg_energy(fit)
```

Excavation Stratigraphic Energy (ESE)

Energy measures local depositional disruption

Dark = stable deposit | Bright = mixing zone (bioturbation, redeposition)

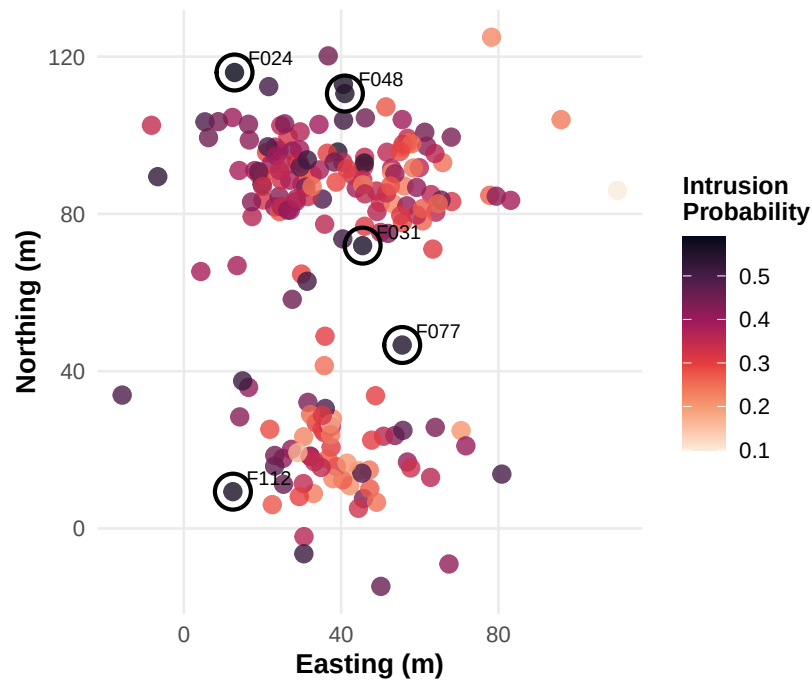


Generated by palimpsestr 0.12.0

```
# Intrusion detection map  
gg_intrusions(fit, top_n = 5)
```

Intrusion Detection Map

19 finds flagged (prob > 0.5) | Circled: top 5 suspects
Score combines high entropy + high energy + low network connectivity

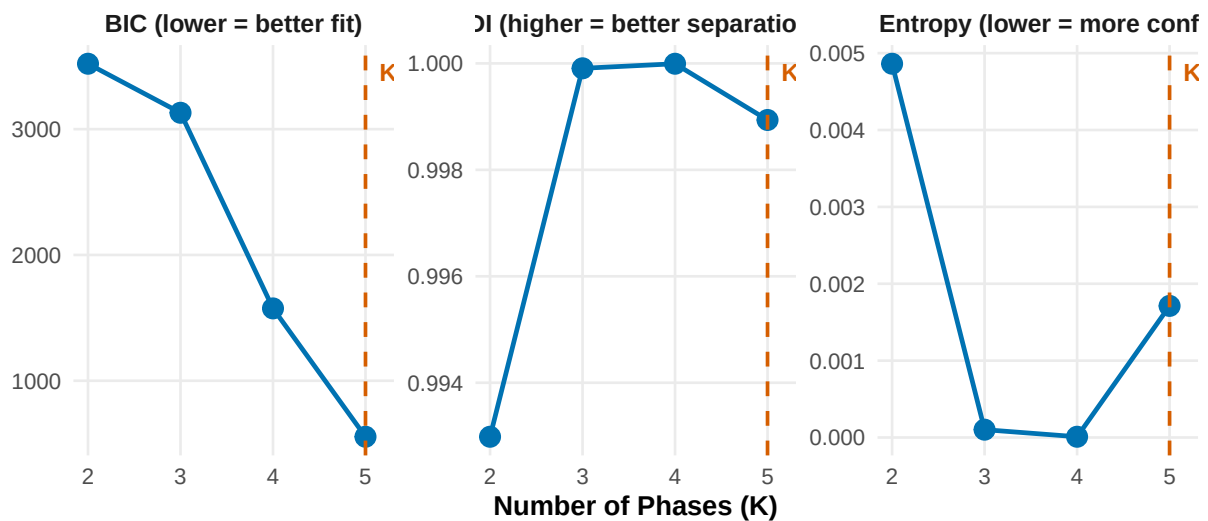


Generated by palimpsestr 0.12.0

```
# Phase count selection diagnostics  
gg_compare_k(ck)
```

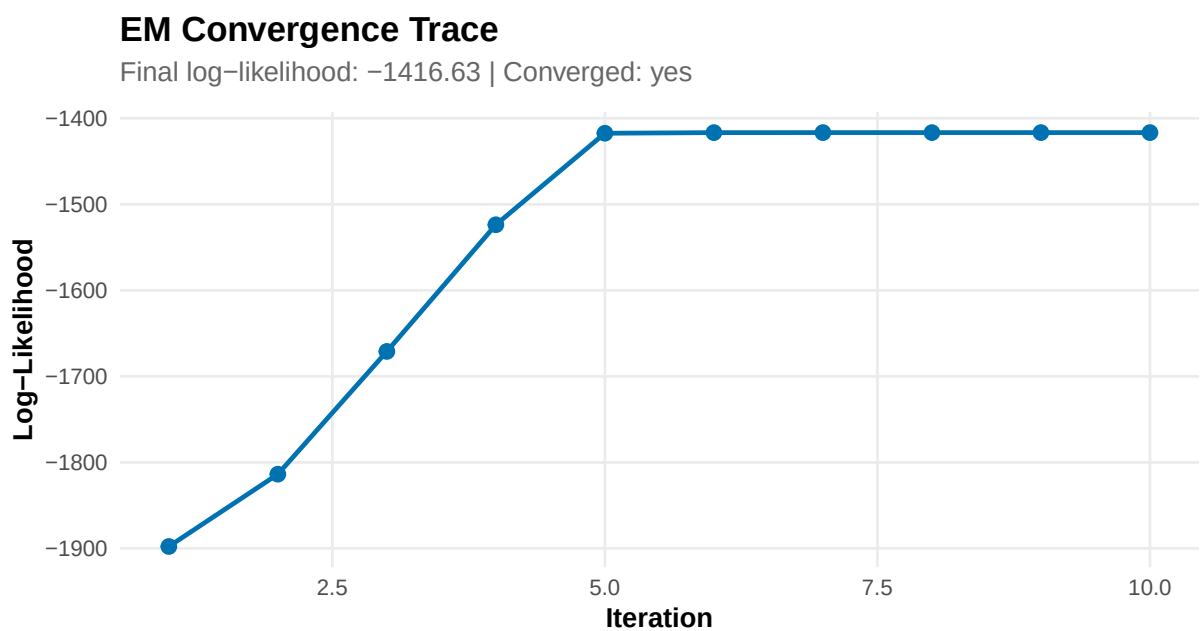
Phase Count Selection

Comparing K = 2 to 5 | Optimal K = 5 (minimum BIC)



Generated by palimpsestr 0.12.0

```
# EM convergence trace  
gg_convergence(fit)
```



Generated by palimpsestr 0.12.0

```
# Vertical phase profile (depth vs horizontal)  
gg_phase_profile(fit)
```


Vertical Phase Profile

Depth (z) vs horizontal position, coloured by phase

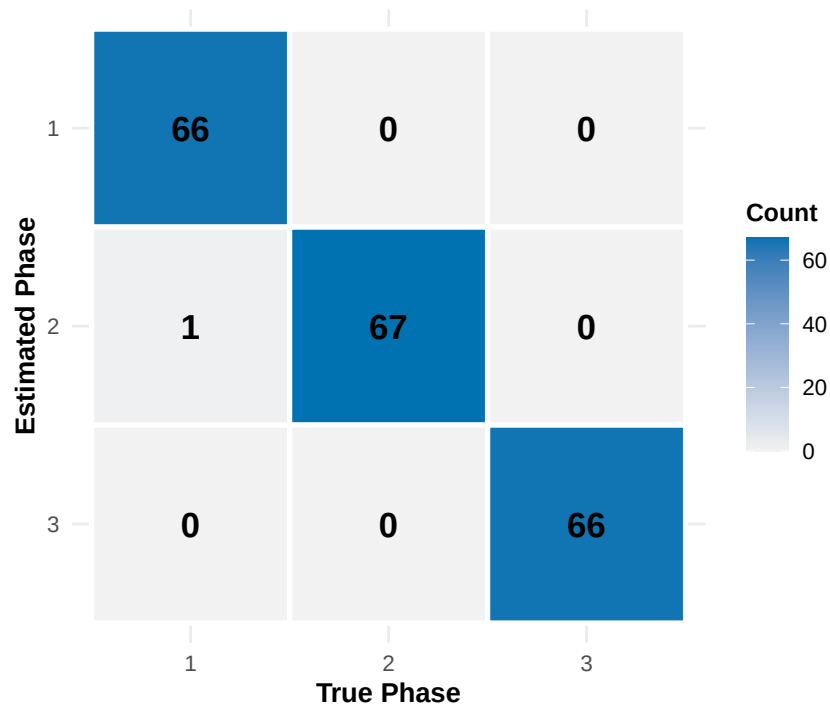


Generated by palimpsestr 0.12.0

```
# Confusion matrix heatmap  
gg_confusion(fit, moderate$true_phase)
```

Phase Confusion Matrix

ARI = 0.985



Generated by palimpsestr 0.12.0

7.2 Interactive plots (plotly)

```
library(plotly)

# Convert any gg_* plot to interactive
as_plotly(gg_phasefield(fit))
as_plotly(gg_entropy(fit))
as_plotly(gg_intrusions(fit))
```

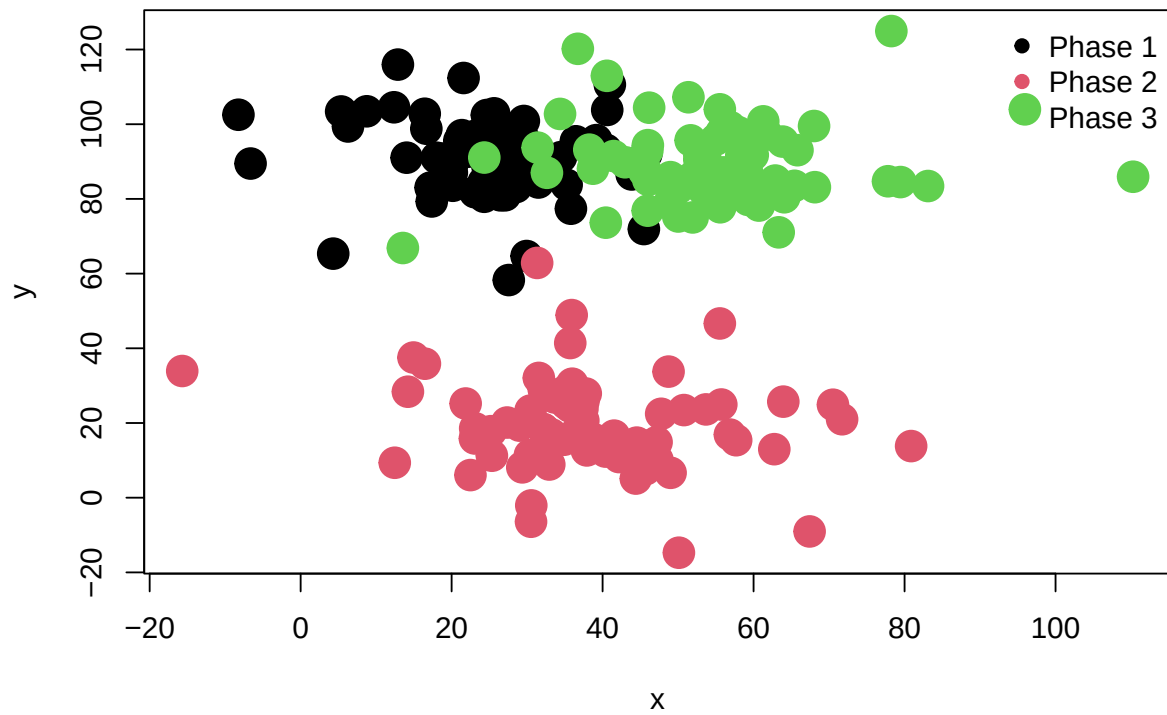
Hover tooltips show: ID, context, phase probability, dating, class, entropy, energy, intrusion score.

7.3 Base R plots

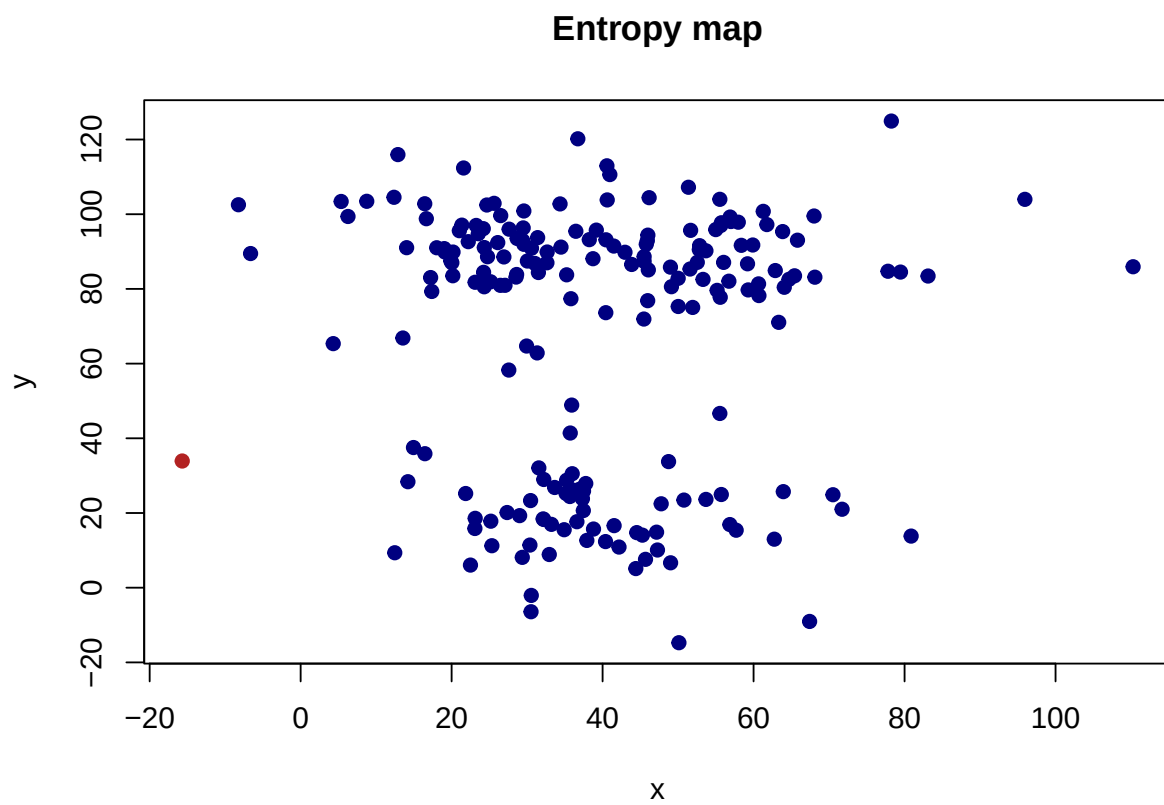
For environments without ggplot2:

```
plot_phasefield(fit)
```

Dominant phase assignment

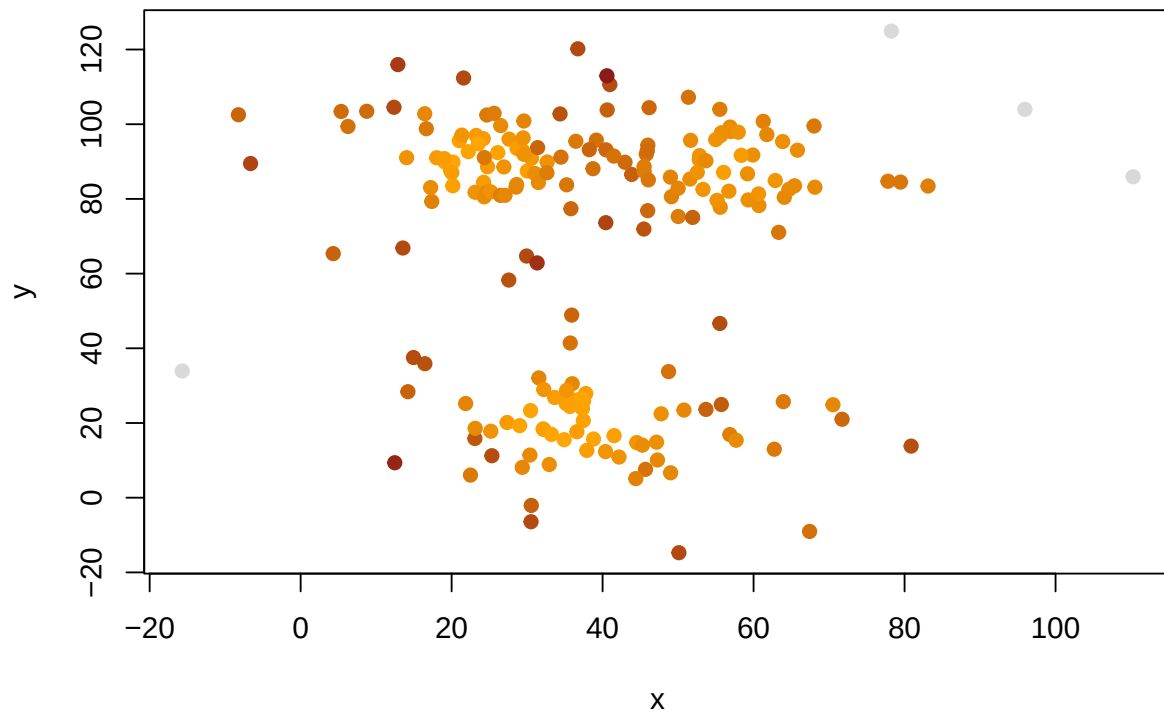


```
plot_entropy(fit)
```



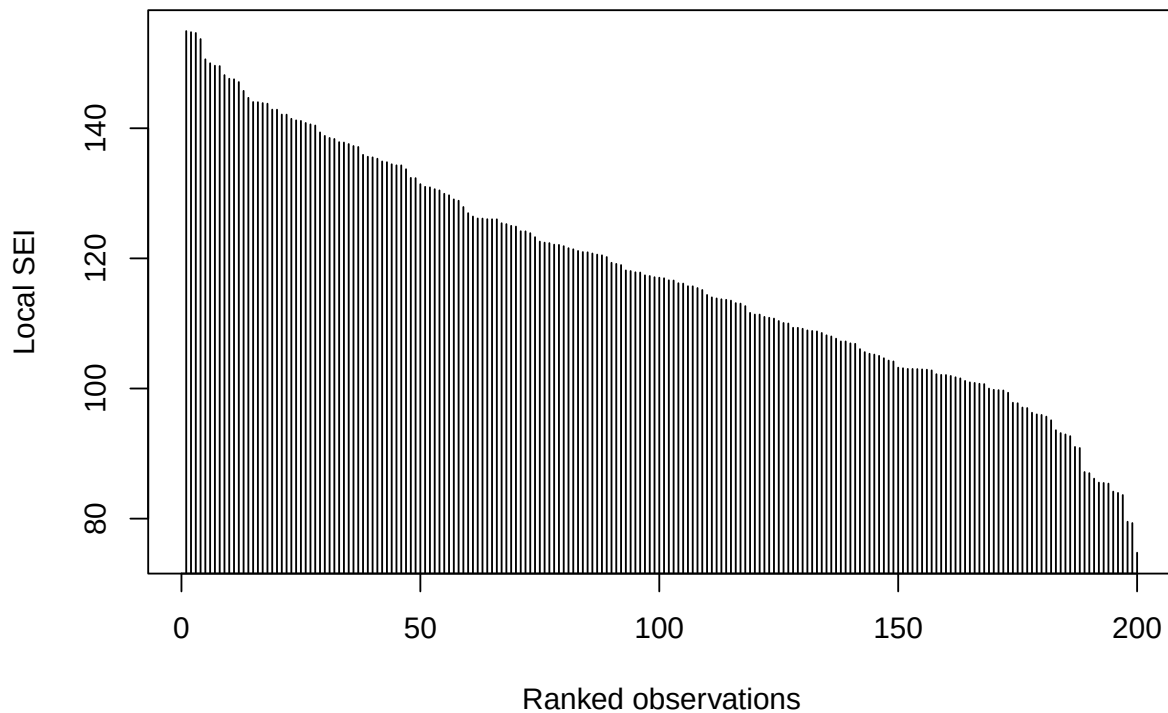
```
plot_energy(fit)
```

Excavation stratigraphic energy



```
plot_sei_profile(fit)
```

Ordered SEI profile



7.4 GIS export

```
library(sf)

# Point layer with phase assignments
sf_pts <- as_sf_phase(fit, crs = 32632L, dims = "XYZ")
st_write(sf_pts, "results.gpkg", layer = "phases")

# High-SEI links as lines
sf_links <- as_sf_links(fit, quantile_threshold = 0.9, crs = 32632L)
st_write(sf_links, "results.gpkg", layer = "sei_links")
```

8 Interpretive Report

Generate a plain-language report for inclusion in excavation documentation:

```
report_sef(fit, lang = "en")
#> # SEF Analysis Report
#>
#> ## Overview
#>
```

```

#> The Stratigraphic Entanglement Field model with K = 3 phases was fitted to a dataset of 200 finds
#>
#> The Palimpsest Dissolution Index (PDI) is 1.000, indicating excellent phase separation. Mean
#>
#> Model diagnostics: LogLik = -1416.6, BIC = 3130.0.
#>
#> ## Phase Composition
#>
#> ### Phase 1 (66 finds, 33.0%)
#>
#> - Dating: average 1001 CE (range: 894 to 1122)
#> - Material classes: bone (13), ceramic (20), lithic (24), metal (9)
#> - Mean entropy: 0.0000; Mean energy: 17.823
#> - Contexts: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - Assessment: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 2 (68 finds, 34.0%)
#>
#> - Dating: average 1146 CE (range: 1004 to 1246)
#> - Material classes: bone (17), ceramic (25), lithic (14), metal (12)
#> - Mean entropy: 0.0003; Mean energy: 17.467
#> - Contexts: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - Assessment: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 3 (66 finds, 33.0%)
#>
#> - Dating: average 1303 CE (range: 1211 to 1393)
#> - Material classes: bone (10), ceramic (22), lithic (20), metal (14)
#> - Mean entropy: 0.0000; Mean energy: 17.364
#> - Contexts: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - Assessment: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ## Intrusion Detection
#>
#> 19 finds (9.5%) have an intrusion probability above 0.5, flagging them as potentially displaced
#>
#> Top suspects:
#>
#> | ID | Context | Intrusion Prob. |
#> |---|---|---|
#> | F024 | SU_1 | 0.590 |
#> | F112 | SU_9 | 0.565 |
#> | F031 | SU_9 | 0.563 |
#> | F048 | SU_2 | 0.558 |
#> | F077 | SU_7 | 0.555 |
#> | F145 | SU_1 | 0.548 |
#> | F028 | SU_3 | 0.545 |
#> | F159 | SU_1 | 0.533 |
#> | F062 | SU_3 | 0.532 |
#> | F125 | SU_6 | 0.518 |
#>
#> ## Stratigraphic Unit Purity
#>

```

```

#> Of 9 stratigraphic units: 0 pure (>90%), 0 mixed (70-90%), 9 palimpsest (<70%).
#>
#> Units classified as palimpsest require special attention, as material from multiple depositional epi.
#>
#> ## Recommendations
#>
#> The deposit shows good phase separation. The model can be used with confidence for phase attribution
#>
#> ---
#> *Report generated by palimpsestr 0.12.0*

```

```

report_sef(fit, lang = "it")
#> # Rapporto Analisi SEF
#>
#> ## Panoramica
#>
#> Il modello Stratigraphic Entanglement Field con K = 3 fasi e' stato applicato a un dataset di 100 reperti.
#>
#> Il Palimpsest Dissolution Index (PDI) e' 1.000, indicando una separazione eccellente delle fasi.
#>
#> Diagnostiche: LogLik = -1416.6, BIC = 3130.0.
#>
#> ## Composizione delle Fasi
#>
#> ### Fase 1 (66 reperti, 33.0%)
#>
#> - Datazione media: 1001 d.C.
#> - Classi materiali: bone (13), ceramic (20), lithic (24), metal (9)
#> - Entropia media: 0.0000; Energia media: 17.823
#> - US: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 totali)
#>
#> ### Fase 2 (68 reperti, 34.0%)
#>
#> - Datazione media: 1146 d.C.
#> - Classi materiali: bone (17), ceramic (25), lithic (14), metal (12)
#> - Entropia media: 0.0003; Energia media: 17.467
#> - US: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 totali)
#>
#> ### Fase 3 (66 reperti, 33.0%)
#>
#> - Datazione media: 1303 d.C.
#> - Classi materiali: bone (10), ceramic (22), lithic (20), metal (14)
#> - Entropia media: 0.0000; Energia media: 17.364
#> - US: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 totali)
#>
#> ## Intrusioni
#>
#> 19 reperti (9.5%) presentano probabilita' di intrusione superiore a 0.5.
#>
#> ## Purezza delle US
#>
#> Su 9 US: 0 pure (>90%), 0 miste (70-90%), 9 palinsesto (<70%).
#>

```



```
#> ---
#> *Rapporto generato da palimpsestr 0.12.0*
```

9 Database Import

Import data from SQLite or PostgreSQL:

```
# SQLite
data <- read_db(
  con = DBI::dbConnect(RSQLite::SQLite(), "excavation.sqlite"),
  table = "finds"
)

# PostgreSQL (PyArchInit)
data <- read_db(
  con = DBI::dbConnect(RPostgres::Postgres(), dbname = "pyarchinit", host = "localhost"),
  table = "finds_table"
)
```

10 Phase Reordering

Mixture models suffer from *label switching*: phase numbers are arbitrary across runs. Use `reorder_phases()` to ensure phase 1 = deepest (oldest) and phase K = shallowest (most recent):

```
fit <- reorder_phases(fit)
table(fit$phase)
#>
#> 1 2 3
#> 66 68 66
```

11 Bootstrap Confidence Intervals

Assess uncertainty on key metrics via bootstrap resampling:

```
bs <- bootstrap_sef(fit, n_boot = 50, verbose = FALSE)
print(bs)
```

#>	statistic	estimate	lower	upper	se
#> 1	pdi	9.999074e-01	9.815071e-01	1.00000000	6.423186e-03
#> 2	mean_entropy	1.017010e-04	1.618468e-10	0.02031658	7.056591e-03
#> 3	mean_energy	1.755018e+01	1.514950e+01	18.22076171	8.190172e-01
#> 4	loglik	-1.416627e+03	-1.420076e+03	984.92508485	9.560619e+02

Returns a data.frame with columns `metric`, `mean`, `se`, `ci_lower`, `ci_upper` for PDI, mean entropy, mean energy, loglik, and optionally ARI (if `true_labels` is provided).

Parameters:

	Parameter	Default	Description
n_boot	100		Number of bootstrap replicates
conf	0.95		Confidence level for intervals
true_labels	NULL		Known true phases for ARI confidence interval
verbose	TRUE		Print progress messages

12 Cross-Validation

Evaluate model fit via k-fold cross-validation:

```
cv <- cv_sef(moderate, k_values = 3, n_folds = 3, seed = 42)
print(cv)
#>  k fold train_loglik test_loglik train_pdi
#> 1 3      1    -896.3563   -471.1466 0.9999998
#> 2 3      2    -260.8705   -204.4386 0.9581275
#> 3 3      3    -780.5878   -369.0116 0.9997784
```

Use with multiple K values to find the best:

```
cv <- cv_sef(data, k_values = 2:5, n_folds = 5, seed = 42)
print(cv)  # shows mean_test_loglik per k
```

13 Data-Driven Weight Optimisation

Instead of manually tuning SEI weights, use `optimize_weights()` to find the best configuration via grid search:

```
opt <- optimize_weights(data, k = 3, n_folds = 3, seed = 42)
print(opt$best_weights)

# Re-fit with optimal weights
fit <- fit_sef(data, k = 3, weights = opt$best_weights)
```

14 Sparse SEI for Large Datasets

For datasets with $n > 1000$, the full $n \times n$ SEI matrix may be too large. Use `sei_sparse()` which automatically applies a distance threshold:

```
# Automatic sparse mode
sei <- sei_sparse(data, max_dist = 5.0)

# Or use sei_matrix() directly with max_dist
sei <- sei_matrix(data, max_dist = 5.0)
```

Pairs beyond `max_dist` Euclidean distance receive SEI = 0, dramatically reducing memory and computation time.

15 Complete Workflow Example

```
library(palimpsestr)

# 1. Load data
data <- read.csv("site_data.csv")

# 2. Select optimal K
ck <- compare_k(data, k_values = 2:6, context = "context",
                 tafonomy = "taf_score", n_init = 3)
gg_compare_k(ck)
best_k <- ck$k[which.min(ck$bic)]

# 3. Fit with optimal K and multiple inits
fit <- fit_sef(data, k = best_k, context = "context",
               tafonomy = "taf_score", n_init = 5, seed = 42)

# 4. Inspect results
print(fit)
gg_phasefield(fit)
gg_entropy(fit)
gg_convergence(fit)
gg_phase_profile(fit)

# 5. Intrusion detection
gg_intrusions(fit, top_n = 10)

# 6. Harris validation
validate_phases_harris(fit)

# 7. Export
export_results(fit, dir = "results/")
report_sef(fit, lang = "en")

# 8. GIS
sf_pts <- as_sf_phase(fit, crs = 32632L)
sf::st_write(sf_pts, "results.gpkg", layer = "phases")
```

16 Real-Data Example: Villa Romana

The `villa_romana` dataset models a multi-period Roman villa excavation with 300 finds across 4 occupation phases (Republican, Early Imperial, Late Imperial, Late Antique) and 18 stratigraphic units. It includes realistic post-depositional disturbance (bioturbation, construction cuts, residual pottery) and taphonomic scoring.

16.1 Load and inspect

```
data(villa_romana)
cat("Finds:", nrow(villa_romana), "| Contexts:", length(unique(villa_romana$context)),
```

```

    "| Classes:", length(unique(villa_romana$class)), "\n")
#> Finds: 615 | Contexts: 54 | Classes: 17
cat("Date range:", min(villa_romana$date_min), "to", max(villa_romana$date_max), "\n")
#> Date range: -650 to 550
table(villa_romana$class)
#>
#>      Campionature      Intonaco Dipinto Materiale Da Costruzione
#>              5              1              38
#>      Materiale Lapideo      Pietre Dure-Gemme      Reperti Antropologici
#>              1              1              1
#>      Reperti Archeozoologici      Reperto Anforaceo      Reperto Ceramico
#>              7              76              417
#>      Reperto Faunistico      Reperto Lapideo      Reperto Metallico
#>              1              6              19
#>      Reperto Numismatico      Reperto Vitreo      Resti Di Pavimentazioni
#>             10             25              1
#>      Scarti Di Produzione      Stucco
#>              5              1

```

16.2 Fit with taphonomic weighting

```

fit_vr <- fit_sef(villa_romana, k = 4,
                 tafonomy = "taf_score",
                 context = "context",
                 n_init = 5, seed = 42)

print(fit_vr)
#> <sef_fit>
#> Observations: 615
#> Estimated phases: 4
#> Mean entropy: 0.000
#> Mean local SEI: 899.803
#> Mean energy: 2.005
#> LogLik: 52469.243
#> BIC: -103789.015
#> ICL: -103789.015
#> PDI: 1.000
#> Converged: yes
#> Initialisations: 5

```

16.3 Diagnostics

```

cat("PDI:", pdi(fit_vr), "\n")
#> PDI: 1
sef_summary(fit_vr)
#> $n
#> [1] 615
#>
#> $k
#> [1] 4

```

```

#>
#> $pdi
#> [1] 1
#>
#> $mean_entropy
#> [1] 3.867719e-10
#>
#> $mean_energy
#> [1] 2.005266
#>
#> $loglik
#> [1] 52469.24
#>
#> $bic
#> [1] -103789
#>
#> $phase_count
#>
#>   1   2   3   4
#> 117  82  25 391

```

16.4 Validation (on simulated data)

Validation with ARI and confusion matrix requires known true phases. We demonstrate with `demo_moderate`:

```

data(demo_moderate)
fit_demo <- fit_sef(demo_moderate, k = 3, seed = 42)
adjusted_rand_index(fit_demo, demo_moderate$true_phase)
#> [1] 1
confusion_matrix(fit_demo, demo_moderate$true_phase)
#>           true
#> estimated  2  1  3
#>           1 67  0  0
#>           2  0 67  0
#>           3  0  0 66

```

16.5 Intrusion detection

```

di_vr <- detect_intrusions(fit_vr)
suspects_vr <- di_vr[di_vr$intrusion_prob > 0.5, ]
cat("Suspected intrusions:", nrow(suspects_vr), "out of", nrow(villa_romana), "\n")
#> Suspected intrusions: 8 out of 615

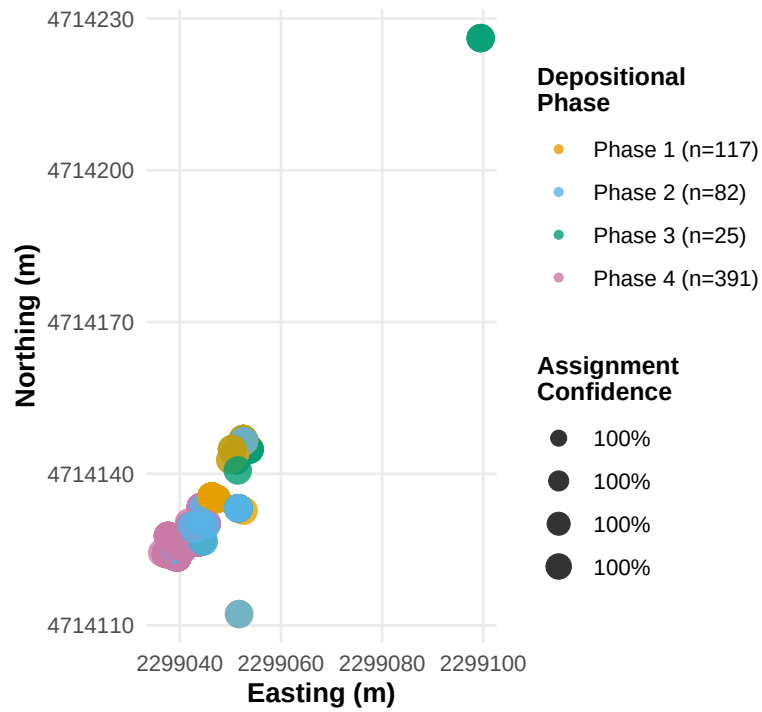
```

16.6 Visualisation

```
gg_phasefield(fit_vr)
```

Dominant Phase Assignment

615 finds assigned to 4 depositional phases | PDI = 1.000

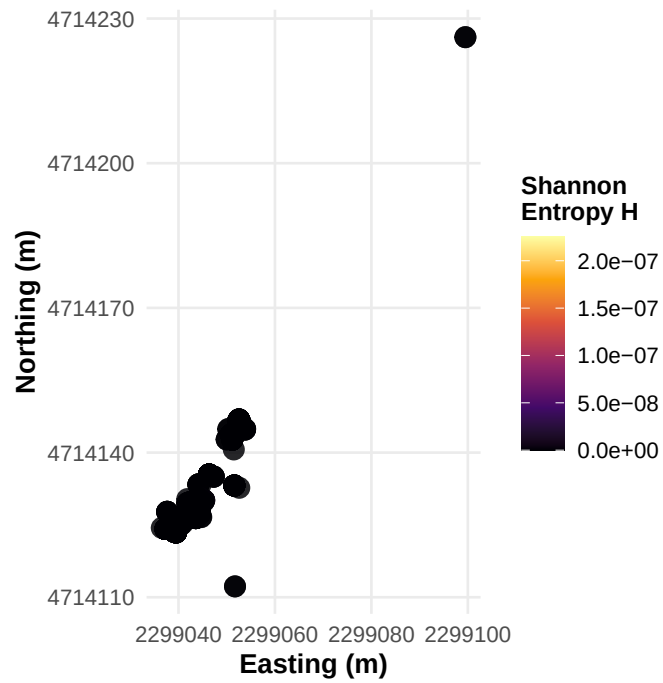


```
gg_entropy(fit_vr)
```

Spatial Entropy Map

Entropy measures uncertainty in phase assignment

Dark = confident assignment | Bright = ambiguous (possible mixing z



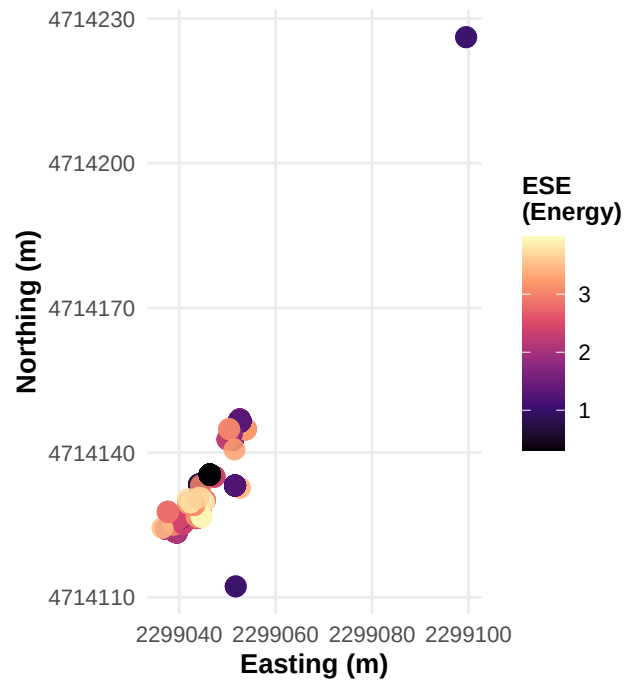
Generated by palimpsestr 0.12.0

```
gg_energy(fit_vr)
```

Excavation Stratigraphic Energy (ESE)

Energy measures local depositional disruption

Dark = stable deposit | Bright = mixing zone (bioturbation, redeposit



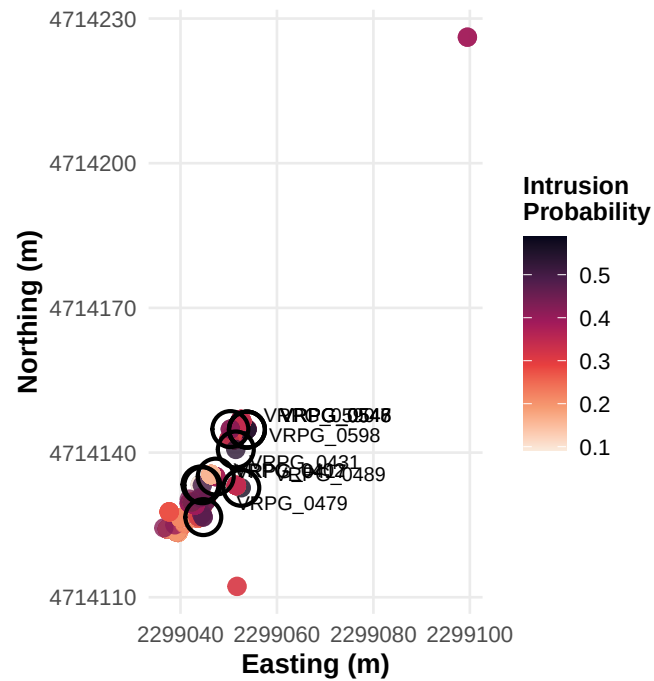
Generated by palimpsestr 0.12.0

```
gg_intrusions(fit_vr, top_n = 10)
```


Intrusion Detection Map

8 finds flagged (prob > 0.5) | Circled: top 10 suspects

Score combines high entropy + high energy + low network connectiv

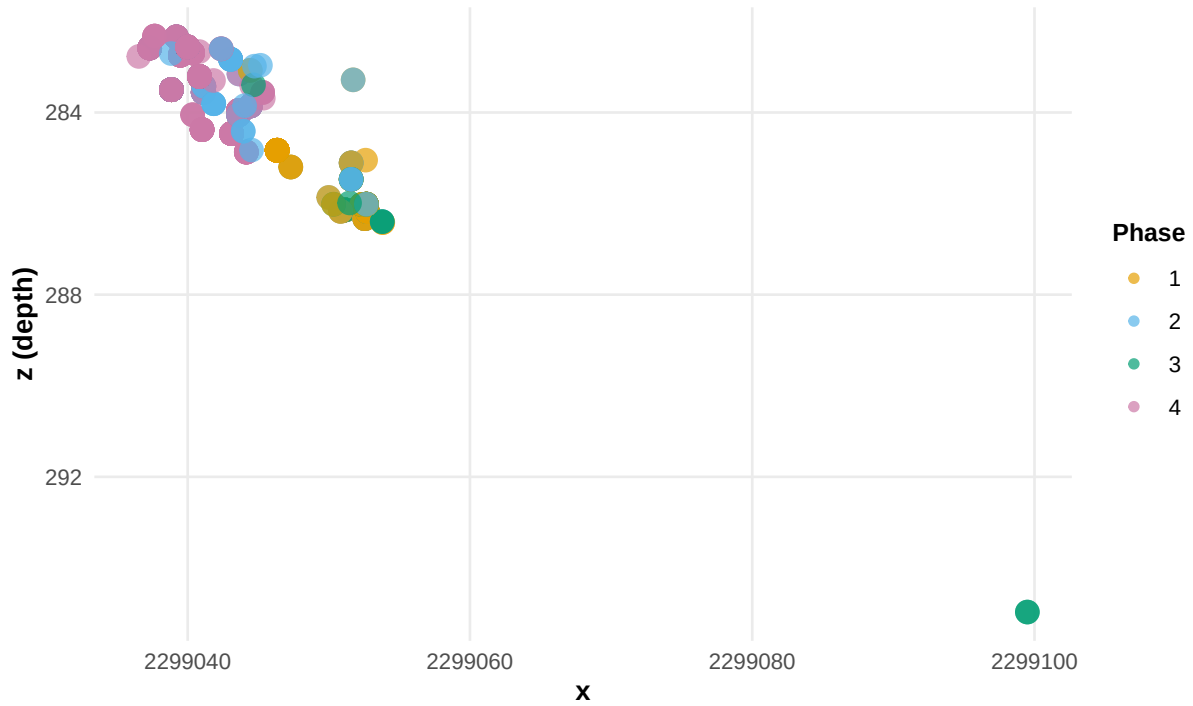


Generated by palimpsestr 0.12.0

```
gg_phase_profile(fit_vr)
```

Vertical Phase Profile

Depth (z) vs horizontal position, coloured by phase



Generated by palimpsestr 0.12.0

16.7 Per-US summary

```
us_summary_table(fit_vr)
#>   context n_finds dominant_phase   purity mean_entropy mean_energy
#> 1  US_102     13           4 1.0000000 2.981066e-17 2.7896559
#> 2  US_104     93           4 0.9139785 2.217938e-12 2.1346706
#> 3  US_105     19           4 1.0000000 4.544040e-21 1.9689254
#> 4  US_108      3           4 1.0000000 2.983874e-18 2.6074504
#> 5  US_111     25           4 0.9600000 9.978852e-12 2.4940144
#> 6  US_114     22           4 0.7727273 9.039273e-19 1.8288814
#> 7  US_116      6           4 0.6666667 3.175876e-12 2.0527589
#> 8  US_117     27           4 0.9629630 4.155994e-12 2.2285513
#> 9  US_118     16           4 0.9375000 2.352234e-11 2.7949156
#> 10 US_133      1           2 1.0000000 5.763193e-27 3.1384414
#> 11 US_134     41           4 0.9268293 1.471072e-12 2.2922646
#> 12 US_136     33           4 1.0000000 5.328291e-11 2.1512502
#> 13 US_137     15           4 0.7333333 1.408285e-10 2.2916569
#> 14 US_144     18           4 0.7222222 1.379959e-10 2.4949012
#> 15 US_148      1           4 1.0000000 1.552959e-20 2.3469451
#> 16 US_149     15           4 1.0000000 1.780079e-12 1.4791983
#> 17 US_150     48           4 0.9583333 1.944366e-12 2.5443835
#> 18 US_151     10           4 1.0000000 2.254134e-08 0.8192432
#> 19 US_157      3           4 1.0000000 1.410648e-14 2.9185184
```

```

#> 20 US_158 1 4 1.0000000 2.962166e-09 2.9116094
#> 21 US_161 4 2 1.0000000 0.000000e+00 3.1492139
#> 22 US_166 2 2 1.0000000 0.000000e+00 2.9669413
#> 23 US_173 1 2 1.0000000 3.929476e-22 3.0153408
#> 24 US_180 1 4 1.0000000 4.547943e-11 3.4764350
#> 25 US_183 5 4 1.0000000 3.154626e-22 2.8218794
#> 26 US_189 2 2 0.5000000 5.727722e-18 3.2728243
#> 27 US_197 3 1 0.6666667 0.000000e+00 3.4870644
#> 28 US_201 1 4 1.0000000 1.636904e-10 3.7261878
#> 29 US_204 4 1 0.7500000 1.979295e-24 2.5724914
#> 30 US_209 1 4 1.0000000 1.443854e-09 3.6537967
#> 31 US_21 2 1 0.5000000 1.421664e-23 1.0000000
#> 32 US_210 41 1 1.0000000 2.118904e-53 0.3773606
#> 33 US_211 1 3 1.0000000 0.000000e+00 3.6575888
#> 34 US_215 1 2 1.0000000 0.000000e+00 3.9862189
#> 35 US_220 1 2 1.0000000 0.000000e+00 3.9065867
#> 36 US_258 8 2 0.8750000 1.415170e-11 3.0820483
#> 37 US_276 1 1 1.0000000 0.000000e+00 3.3655781
#> 38 US_287 25 1 0.4800000 8.608797e-21 1.3064744
#> 39 US_296 10 1 0.7000000 4.270786e-20 1.4896941
#> 40 US_297 17 1 0.7647059 1.906397e-20 1.1256531
#> 41 US_299 1 1 1.0000000 0.000000e+00 2.7606762
#> 42 US_302 2 3 1.0000000 1.872049e-39 1.0000000
#> 43 US_303 1 1 1.0000000 0.000000e+00 2.7321978
#> 44 US_304 3 3 0.6666667 9.585193e-21 3.1860488
#> 45 US_306 5 1 0.6000000 0.000000e+00 1.5398238
#> 46 US_307 30 1 0.6333333 2.659448e-20 1.1739085
#> 47 US_308 4 1 0.7500000 1.993070e-21 1.9313491
#> 48 US_315 2 1 0.5000000 2.006997e-21 2.2944214
#> 49 US_316 2 1 0.5000000 0.000000e+00 3.1262316
#> 50 US_319 6 1 0.5000000 7.969753e-21 1.0976667
#> 51 US_324 1 3 1.0000000 0.000000e+00 3.2689100
#> 52 US_72 5 4 0.8000000 4.489460e-11 3.5532633
#> 53 US_76 11 1 0.4545455 2.207806e-21 1.1063421
#> 54 US_89 1 2 1.0000000 0.000000e+00 3.6314873
#> mean_local_sei n_intrusions
#> 1 1072.3812 0
#> 2 1167.1218 0
#> 3 1126.0732 0
#> 4 1083.7532 0
#> 5 1001.6875 0
#> 6 1069.2456 0
#> 7 959.4754 0
#> 8 1017.5074 0
#> 9 1015.1978 0
#> 10 726.3723 0
#> 11 1108.6761 0
#> 12 1037.5226 0
#> 13 853.6961 0
#> 14 919.1431 0
#> 15 1103.1060 0
#> 16 1072.9793 0
#> 17 1081.4370 0

```

```

#> 18      1013.2769      1
#> 19      1140.6567      0
#> 20       798.1423      0
#> 21       733.7632      0
#> 22       667.3538      0
#> 23       335.7572      0
#> 24       769.7212      0
#> 25      1069.4656      0
#> 26       924.1481      0
#> 27       713.2222      0
#> 28       803.9810      0
#> 29       478.5477      1
#> 30       803.3619      0
#> 31       511.7135      0
#> 32       648.7849      0
#> 33       733.2766      0
#> 34       722.3579      0
#> 35       708.2789      0
#> 36       744.1272      0
#> 37       188.3032      1
#> 38       447.8803      0
#> 39       495.4881      0
#> 40       544.4120      0
#> 41       666.3856      0
#> 42        74.1208      0
#> 43       665.8930      0
#> 44       270.7079      3
#> 45       369.1143      0
#> 46       473.5014      0
#> 47       396.6293      0
#> 48       501.2748      0
#> 49       485.3525      1
#> 50       542.3492      0
#> 51       276.6555      1
#> 52       894.3902      0
#> 53       436.7971      0
#> 54       702.3748      0

```

16.8 Interpretive report

```

report_sef(fit_vr, lang = "en")
#> # SEF Analysis Report
#>
#> ## Overview
#>
#> The Stratigraphic Entanglement Field model with K = 4 phases was fitted to a dataset of 615 fi
#>
#> The Palimpsest Dissolution Index (PDI) is 1.000, indicating excellent phase separation. Mean
#>
#> Model diagnostics: LogLik = 52469.2, BIC = -103789.0.
#>

```

```

#> ## Phase Composition
#>
#> ### Phase 1 (117 finds, 19.0%)
#>
#> - **Dating**: average 69 CE (range: -650 to 550)
#> - **Material classes**: Materiale Da Costruzione (1), Reperti Archeozoologici (7), Reperto Anforaceo
#> - **Mean entropy**: 0.0000; **Mean energy**: 1.037
#> - **Contexts**: US_197, US_204, US_21, US_210, US_276, US_287, ... (17 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 2 (82 finds, 13.3%)
#>
#> - **Dating**: average 413 CE (range: 138 to 550)
#> - **Material classes**: Materiale Da Costruzione (36), Reperto Anforaceo (11), Reperto Numismatico (
#> - **Mean entropy**: 0.0000; **Mean energy**: 2.436
#> - **Contexts**: US_104, US_111, US_114, US_116, US_117, US_118, ... (32 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 3 (25 finds, 4.1%)
#>
#> - **Dating**: average 367 CE (range: 180 to 450)
#> - **Material classes**: Campionature (5), Intonaco Dipinto (1), Materiale Da Costruzione (1), Materi
#> - **Mean entropy**: 0.0000; **Mean energy**: 1.748
#> - **Contexts**: US_211, US_287, US_297, US_302, US_304, US_306, ... (10 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 4 (391 finds, 63.6%)
#>
#> - **Dating**: average 438 CE (range: 425 to 450)
#> - **Material classes**: Reperto Anforaceo (63), Reperto Ceramico (328)
#> - **Mean entropy**: 0.0000; **Mean energy**: 2.221
#> - **Contexts**: US_102, US_104, US_105, US_108, US_111, US_114, ... (26 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ## Intrusion Detection
#>
#> **8 finds** (1.3%) have an intrusion probability above 0.5, flagging them as potentially displaced o
#>
#> Top suspects:
#>
#> | ID | Context | Intrusion Prob. |
#> |---|---|---|
#> | VRPG_0402 | US_151 | 0.588 |
#> | VRPG_0489 | US_276 | 0.573 |
#> | VRPG_0431 | US_204 | 0.559 |
#> | VRPG_0590 | US_316 | 0.539 |
#> | VRPG_0598 | US_324 | 0.539 |
#> | VRPG_0548 | US_304 | 0.538 |
#> | VRPG_0547 | US_304 | 0.537 |
#> | VRPG_0546 | US_304 | 0.525 |
#>
#> ## Stratigraphic Unit Purity
#>

```

```
#> Of 54 stratigraphic units: 33 pure (>90%), 9 mixed (70–90%), 12 palimpsest (<70%).
#>
#> Units classified as palimpsest require special attention, as material from multiple depositional epi.
#>
#> ## Recommendations
#>
#> The deposit shows good phase separation. The model can be used with confidence for phase attribution.
#>
#> ---
#> *Report generated by palimpsestr 0.12.0*
```

17 Function Reference

17.1 Core

Function	Description
<code>fit_sef()</code>	Fit the SEF model
<code>compare_k()</code>	Compare multiple K values
<code>archaeo_sim()</code>	Simulate archaeological data
<code>reorder_phases()</code>	Relabel phases by mean depth (deepest = phase 1)

17.2 Diagnostics

Function	Description
<code>pdi()</code>	Palimpsest Dissolution Index
<code>sef_summary()</code>	Compact model summary
<code>as_phase_table()</code>	Phase probability table
<code>predict_phase()</code>	Alias for <code>as_phase_table()</code>
<code>phase_diagnostic_table()</code>	Full diagnostic table
<code>detect_intrusions()</code>	Intrusion detection

17.3 Validation

Function	Description
<code>adjusted_rand_index()</code>	Adjusted Rand Index
<code>confusion_matrix()</code>	Confusion matrix (auto-reordered)

17.4 Export

Function	Description
<code>us_summary_table()</code>	Per-US summary statistics
<code>phase_transition_matrix()</code>	Vertical transition counts
<code>export_results()</code>	Export all results to CSV

17.5 Harris Matrix

Function	Description
<code>harris_from_contexts()</code>	Auto-generate penalty from depths
<code>read_harris()</code>	Import Harris Matrix from CSV
<code>validate_phases_harris()</code>	Check phase-stratigraphy consistency

17.6 Robustness & Sensitivity

Function	Description
<code>bootstrap_sef()</code>	Bootstrap confidence intervals on PDI, entropy, ARI
<code>cv_sef()</code>	K-fold cross-validation for model selection
<code>optimize_weights()</code>	Data-driven SEI weight optimisation via grid search

17.7 SEI / ESE

Function	Description
<code>sei_matrix()</code>	Stratigraphic Entanglement Index matrix
<code>sei_sparse()</code>	Sparse SEI for large datasets ($n > 1000$)
<code>local_sei()</code>	Row sums of SEI matrix
<code>ese()</code>	Excavation Stratigraphic Energy

17.8 Plots (ggplot2)

Function	Description
<code>gg_phasefield()</code>	Phase assignment map
<code>gg_entropy()</code>	Entropy map
<code>gg_energy()</code>	ESE energy map
<code>gg_intrusions()</code>	Intrusion detection map
<code>gg_compare_k()</code>	K selection diagnostics
<code>gg_convergence()</code>	EM convergence trace
<code>gg_phase_profile()</code>	Vertical depth profile
<code>gg_confusion()</code>	Confusion matrix heatmap
<code>gg_cv()</code>	Cross-validation diagnostic plot
<code>gg_bootstrap()</code>	Bootstrap confidence interval plot
<code>gg_weights()</code>	Weight sensitivity heatmap
<code>gg_map()</code>	Overlay on excavation plan
<code>as_plotly()</code>	Convert any gg_* to interactive

17.9 Plots (base R)

Function	Description
<code>plot_phasefield()</code>	Phase map
<code>plot_entropy()</code>	Entropy map
<code>plot_energy()</code>	Energy map
<code>plot_sei_profile()</code>	SEI profile

17.10 GIS

Function	Description
<code>as_sf_phase()</code>	Export as sf point layer
<code>as_sf_links()</code>	Export SEI links as sf lines
<code>load_geometries()</code>	Load excavation plan geometries

17.11 Other

Function	Description
<code>report_sef()</code>	Interpretive report (en/it)
<code>read_db()</code>	Import from SQLite/PostgreSQL